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A multisource dataset for anomaly detection and fault prediction in urban water distribution networks

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Abstract

This multi-source dataset was compiled to support research on anomaly-based leak detection in urban water distribution networks (WDNs). It contains one year of hourly data collected from a Slovak water utility, combining supervisory control and data acquisition (SCADA) measurements (flow and pressure), energy consumption variables (kilowatts, kW; kilovolt-ampere reactive, kvar), and environmental indicators such as groundwater level, temperature, and humidity. All features were transformed into standardized anomaly scores on a 0–100 scale using Elastic Machine Learning and Isolation Forest methods. Confirmed leak records from the utility's operational information system were mapped to binary labels using a ± 7 -day temporal window. Feature selection resulted in 18 variables retained based on their statistical association with leak labels using the Goodman–Kruskal gamma coefficient. The dataset can be used for benchmarking anomaly detection and prediction models, evaluating lead-time sensitivity, and developing data-driven early-warning systems. It is also suitable for studies on spatially segmented WDN analysis and is publicly available to support reproducible research.

Background & Summary

Urban water distribution networks (WDNs) are essential infrastructure systems that transport treated water from natural sources (rivers, lakes, or groundwater) to residential, commercial, and industrial users. They comprise interconnected pipes, pumps, valves, storage tanks, and treatment facilities that must operate reliably to maintain adequate flow, pressure, and water quality.

A long-standing challenge in operating WDNs is the presence of non-revenue water (NRW), defined as water that enters the distribution system but does not result in billed consumption due to physical losses (encompassing leaks), commercial losses (theft/meter errors), or authorized but unbilled use (water allocated for firefighting, public parks, or system flushing that is not charged to customers) [1,2].

Globally, NRW levels are commonly estimated at 25–35% [3], creating significant economic, operational, and environmental burdens for water utilities [4,5].

Leakage is a major component of physical losses and is traditionally addressed through methods such as acoustic and visual inspection (limited to surface leaks), pressure (insufficient for detecting small leaks) and flow monitoring, step testing (the temporary shutdown of network sections, leading to service interruptions) and tracer-based techniques (effective for localized leak detection rather than large-scale networks) [6]. While effective in localized or short-term applications, these approaches often fail to identify small or unreported leaks that can persist over long periods [7].

In conventional WDNs, leak detection depends on manual inspection, acoustic devices, and pressure monitoring, while modern WDNs utilize smart sensors, AI-driven detection, real-time monitoring, and satellite imaging. Similarly, data collection in conventional WDNs is based on manual logging and periodic field surveys, whereas in modern WDNs, data collection is conducted through SCADA, IoT, and cloud-

based technologies. The increasing complexity and heterogeneity of modern WDNs have challenged traditional monitoring systems, prompting a shift toward data-driven methodologies that integrate internal telemetry with external environmental data [8,9].

To support such research, this dataset was created in collaboration with a Slovak water utility as a pilot example of multi-source monitoring in a real WDN. It provides one year of hourly hydraulic, energy, and environmental data for a single distribution zone, along with temporally aligned leak records. Eighteen variables were retained based on statistical association with leak labels, enabling reuse for benchmarking anomaly detection models, lead-time analyses, early-warning system development, and studies of spatially segmented WDNs [7]. The dataset is openly available to promote reproducible research and future comparative studies.

Methods

Dataset description and design

The dataset was constructed to support research on anomaly-based leak detection in WDNs by integrating heterogeneous operational and environmental time series [10].

It consists of curated, harmonized, and annotated hourly data collected over one year (2022) from a single WDN zone operated by a Slovak water utility [10]. The design follows established practices for infrastructure telemetry analysis and reproducible data preparation.

To ensure transparent, reproducible, and methodologically consistent data preparation, we implemented a structured, multi-stage data processing workflow. The process, summarized in Fig. 1, outlines the sequential logic behind the dataset construction.

Figure 1 goes here (Fig1.jpg).

Selection of the WDN zone

The study WDN zone was selected in collaboration with utility engineers based on predefined operational criteria, including hydraulic isolation, sensor availability, fault-report history, infrastructure heterogeneity, and the presence of both urban and peri-urban characteristics. The selected zone includes multiple municipalities, pumping stations, water sources, and metering points, and represents a mix of newer and legacy infrastructure, providing a realistic operational landscape for evaluating data-driven leak detection techniques. The zone was favored for its extensive sensorization, including internal telemetry (flow, pressure, temperature, discharge vigor, and energy consumption) and externally correlated indicators (e.g., groundwater sensors and meteorological stations).

Ground-level utility staff identified this segment as particularly prone to minor, recurring, and often undetected leaks - especially those that do not manifest through sudden flow anomalies or visible bursts [7]. Sensor locations were georeferenced using internal GIS tools and abstracted via polygon-based geofencing (Fig. 2) to preserve operational confidentiality while ensuring spatial consistency [7].

Figure 2 goes here (Fig2.jpg).

Although the hydraulic, energy, and environmental indicators were analyzed separately in initial stages, this was done to quantify their standalone predictive value before combining them into a unified dataset for integrated modeling.

Data acquisition

Data were collected from three primary domains [11-13]:

1. **Internal SCADA telemetry:** hourly *flow rate*, *pressure*, *water source level*, *discharge*, and *temperature*.
2. **Energy telemetry:** *active power (kW)*, *reactive power (kvar)*, and *hourly work (kW-hour)*.
3. **Environmental context:** groundwater levels, groundwater temperature, precipitation, humidity, and surface temperature.

SCADA and energy data were accessed under institutional agreement, while environmental inputs were sourced from openly available repositories. Only derived and anonymized indicators are included in the public release.

Data integration and preprocessing

All data sources were integrated using a PostgreSQL-based Extract–Transform–Load (ETL) pipeline consistent with the Cross-Industry Standard Process for Data Mining (CRISP-DM) methodology [14]. Heterogeneous formats (SQL databases, APIs, CSV, JSON) were normalized, temporally aligned to an hourly resolution, and unified within a common time index. Sensors with less than 70% annual data availability were excluded. Invalid placeholder values were replaced with nulls, and no interpolation or gap filling was applied.

Variables were organized using a two-level classification system:

1. **Origin-based classification:** distinguishing between internal and external variables.
 - **Internal Variables** (collected within the operational boundaries of the WDN), primarily derived from SCADA telemetry and energy metering systems [15,16]:
 - **Traditional:** *Flow* and *pressure* sensors,
 - **Non-traditional:** *Water source temperature*, *level*, and *discharge* vigor; *Energy use* indicators such as *active/reactive power*, *hourly energy metrics*.
 - **External variables** (acquired from environmental or meteorological systems, external sources to the WDN):
 - *Groundwater level* and *temperature* (e.g., *gw_lvl_*, *gw_temp_*),
 - *Surface temperature*, *humidity*, *precipitation*, *river level* and *flow*, *snow cover* (non-traditional).
2. **Relevance-based classification:** identifying variables as traditional or non-traditional in leak detection frameworks.

The second classification layer groups variables based on domain relevance:

- **Traditional variables** include only:
 - *Flow rate* (e.g., *_ws_flow*),
 - *Pressure* (e.g., *_ws_pressure*).

These are standard metrics used in most real-time control systems [6,7]. However, in this dataset, their predictive power was weak ($\gamma < 0.1$).

- **Non-traditional variables** encompass:
 - *All energy-related features* (e.g., *_kW*, *_power_hour*, *_kvar_hour*),
 - *Environmental indicators* (e.g., *temperature*, *humidity*, *groundwater conditions*),
 - *Water source-specific metrics* (e.g., *_ws_temp*, *_ws_vigor*, *_ws_level*).

This two-dimensional structure facilitates systematic reuses and comparative model evaluation. Several external non-traditional variables showed stronger associations with leak events, highlighting the importance of integrating external environmental data into internal WDN monitoring [9].

A central *hourly frame* was established to integrate sensor readings, ensuring consistency across data sources. Sensors with annual uptime below 70% were excluded to prevent bias from sparse coverage.

Placeholder values such as -9999 or 9999 (indicating sensor faults or missing data) were replaced with NULLs. No interpolation or artificial gap filling was applied, preserving the temporal integrity required for time-series anomaly analysis and model training.

Anomaly score generation

Raw time-series measurements – original sensor values expressed in their native units (e.g., kW, °C, or m³/h) prior to any transformation – were transformed into standardized anomaly scores ranging from 0 to 100 to enable cross-variable comparability (0 represents a typical or expected reading and 100 corresponds to a statistically extreme or rare value based on historical behaviours).

Internal telemetry variables were processed using machine learning (ElasticSearch ML, X-Pack) configured with hourly resolution and daily and weekly seasonality [17]. It utilizes a single-metric job type with 24-hour periodicity and enabled daily and weekly seasonality. It is particularly well-suited for infrastructure telemetry that exhibits regular operational cycles and is optimized for single-sensor streams [18,19].

Figure 3 goes here (Fig3.jpg).

Fig. 3 presents a real-world anomaly detection case recorded by Elastic ML for one of the internal telemetry variables-likely an energy consumption metric such as active power (kW) at a WDN pump station. The sharp spike, highlighted by a red dot, significantly deviates the modelled seasonal baseline and is classified as a critical anomaly. The shaded blue areas denote the model's expected range, while the lower bar summarizes the severity and duration of all detected anomalies over the monitored period. The timing and magnitude of this peak align closely with a confirmed leak event, demonstrating the effectiveness of real-time anomaly scoring in providing early-warnings within infrastructure systems [9]. External environmental variables were processed using the Isolation Forest algorithm implemented in scikit-learn. The Isolation Forest model was configured with $n_estimators = 100$, $max_features = 1.0$, $contamination = 0.05$, and $random_state = 42$. These settings follow standard best practices and were tuned to the hourly periodicity of the dataset. This method is well-suited for identifying outliers in multidimensional feature spaces, capturing complex and contextual patterns in groundwater temperature, humidity, and meteorological signals [9].

Anomaly scores were validated against real-world leak labels using statistical association tests. Notably, several environmental anomaly scores (e.g., groundwater level and temperature) demonstrated strong associations with *fault_d7* leak indicators, achieving high Goodman–Kruskal gamma coefficients ($\gamma > 0.34$, $p < 0.001$). Similarly, internal energy-related anomaly scores (e.g., *_power_hour*, *_kvar_hour*) showed significant predictive strength (power).

Conversely, flow- and pressure-based anomaly scores - despite being central in traditional leak detection - showed negligible statistical association with reported faults ($\gamma < 0.1$, $p > 0.1$). This finding underscores the limited predictive value of these indicators in early warning applications without contextual enrichment [7].

These observations highlight the relevance of combining statistical time-series modeling with operational domain knowledge within this dataset, demonstrating how multi-source data can be applied for exploratory validation of anomaly detection techniques [15,17].

Both approaches were selected as complementary algorithms for anomaly detection. Elastic ML performs probabilistic seasonal modeling suitable for structured telemetry with regular cycles, while Isolation Forest identifies outliers in multidimensional environmental data. This combination ensures robust detection across heterogeneous data domains.

Fault labeling and temporal window selection

Leak events were identified using confirmed maintenance records from the utility's operational information system (OIS), which documents validated maintenance reports linked to physical pipeline faults. These records were used to generate binary classification labels at an hourly resolution. The primary label, denoted *fault_d7*, indicates whether a given record falls within a ± 7 -day window surrounding a confirmed leak event.

The ± 7 -day window was selected through an empirical time-shift analysis informed by practice in time-series anomaly detection and infrastructure monitoring. Statistical evaluations were conducted using Goodman–Kruskal gamma coefficients and generalized linear models across multiple variables and time offsets. This analysis revealed that anomalous patterns in several non-traditional variables (e.g., groundwater level, energy consumption, and discharge-related indicators) frequently appeared days before or persisted after the official leak report date, reflecting uncertainty in detection and repair timing in operational settings [9].

Fig. 4 presents the gamma coefficient as a function of time-shift. The plot displays mean gamma values (points) for each time-shift condition (*fault* to *fault_d10*), with error bars representing standard deviation ($\pm$ SD). A clear upward trend in gamma is observed with increasing time-shifts. However, for the last four time-shifts (*fault_d7* to *fault_d10*), the mean gamma values no longer show statistically significant increases. In contrast, the variability for the last three time-shifts (*fault_d8* to *fault_d10*) increases significantly. Post-hoc analysis using Bonferroni test confirmed that the last four time-shifts (*fault_d7* to *fault_d10*) form a homogeneous group ($p > 0.05$), indicating that beyond a ± 7 -day time-shift, the gamma coefficient does not significantly increase.

Figure 4 goes here (Fig4.jpg).

To evaluate temporal sensitivity, auxiliary labels representing ± 1 to ± 10 -day windows surrounding leak events (*fault_d1* to *fault_d10*) were generated for internal analysis. The results confirmed that detection performance plateaus at the ± 7 -day window, consistent with best practices in condition-based monitoring and early-warning systems. As a result, the *fault_d7* was selected as the most statistically robust and operationally relevant labeling strategy, optimizing early detection capability with practical fault-reporting timelines [9].

Variable filtering and statistical validation

Following anomaly score generation, a systematic variable filtering procedure was applied to reduce model dimensionality and retain statistically relevant features. The initial feature set comprised more than 50 anomaly-transformed time-series derived from both internal SCADA, energy, and external environmental data sources. Each variable was evaluated using the Goodman-Kruskal gamma coefficient to quantify its ordinal association with the binary *fault_d7* label.

Variables were retained if they met two criteria: 1) statistical significance ($p < 0.05$) and 2) minimum association threshold ($\text{gamma} \geq 0.10$). This approach balances effect size and statistical confidence, which is particularly crucial for noisy infrastructure telemetry data[6,7].

Eighteen variables satisfied these criteria (Tab. 1) and were retained in the final dataset. These included both internal operational indicators and external environmental variables, demonstrating their predictive strength in relation to leak events.

These variables represent anonymized sensor-based anomaly score (0–100) rather than discrete fault events. Traditional hydraulic indicators (flow and pressure) did not meet the selection thresholds and were excluded, as their associations with the fault label were not statistically significant within this dataset.

While these findings highlight the potential of non-traditional variables, they should be interpreted as dataset-specific and exploratory, reflecting the characteristics of the studied WDN zone. Broader validation across additional utilities, operational contexts, and multi-year datasets will be essential to confirm the generalizability of these observations.

Table 1. List of 18 sensor-level anomaly score variables with statistically significant association ($\gamma \geq 0.10$, $p < 0.05$) to confirmed leak events (*fault_d7*). Numeric identifiers denote anonymized sensor IDs.

Variable Name	Category	Gamma (γ)	p-value	Association Strength
4700008966_kW	Energy	+0.25	< 0.001	Low to Moderate
4700008966_power_hour	Energy	+0.25	< 0.001	Low to Moderate
4408411600_kW	Energy	+0.16	< 0.05	Low
4408411600_power_hour	Energy	+0.14	< 0.01	Low
4608927500_kvar_hour	Energy	+0.13	< 0.05	Low
238045_ws_temp	Water Source	+0.25	< 0.01	Low to Moderate
279625_ws_temp	Water Source	+0.25	< 0.001	Low to Moderate
279804_ws_level	Water Source	+0.27	< 0.001	Low to Moderate
279805_ws_vigor	Water Source	+0.27	< 0.001	Low to Moderate
279623_ws_vigor	Water Source	+0.33	< 0.001	Moderate
319235_ws_temp	Water Source	+0.38	< 0.001	Moderate
279831_ws_temp	Water Source	+0.27	< 0.001	Low to Moderate
326417_ws_temp	Water Source	+0.34	< 0.001	Moderate
326416_ws_vigor	Water Source	+0.12	< 0.001	Low
temp_site1_anomaly_score	Environmental	+0.18	< 0.001	Low
sfc_temp_site1_anomaly_score	Environmental	+0.14	< 0.001	Low
gw_lvl_site2_anomaly_score	Environmental	+0.44	< 0.001	Moderate to Strong
gw_temp_site2_anomaly_score	Environmental	+0.34	< 0.001	Moderate

Traditional hydraulic indicators (*flow_ws* and *pressure_ws*) did not meet the selection thresholds and were excluded, as their associations with the fault label were not statistically significant within this dataset ($\gamma = -0.06$ for *flow*; $\gamma = -0.26$ for *pressure*, both $p > 0.1$). While flow and pressure are foundational to hydraulic analysis, our study found that several leak events were statistically more strongly associated with energy or environmental anomaly signals than with hydraulic variables alone. [7].

To support interpretability, anomaly score trajectories were visually inspected across internal operational, energy, and environmental domains relative to the *fault_d7* windows (Fig. 5).

Figure 5 goes here (Fig5.jpg).

Anomaly pattern verification

Red-shaded areas (Fig. 5) denote leak proximity intervals (*fault_d7*). These visualizations were used to verify temporal alignment between anomaly patterns and labeled fault intervals and to assess differences in anomaly structure across data categories and detection methods.

The time-aligned visualization reveals that anomaly score spikes frequently occur within ± 7 -day windows of confirmed leak events (*fault_d7*). This alignment is particularly strong in energy-related signals, such as *4700008966_power_hour* and *4408411600_kW*, where narrow, high-amplitude peaks coincide with fault labels, confirming the model's ability to capture pre-failure operational stress.

Internal telemetry and energy-related variables processed using Elastic ML typically exhibited sparse but high-amplitude anomaly peaks, while environmental variables processed using Isolation Forest showed broader temporal clustering of moderate anomaly scores. These complementary behaviors reflect differences in data structure and detection algorithms and informed the inclusion of heterogeneous variables in the final dataset [17].

Together, these visual confirmations validate both the model architecture and the fault labeling approach (± 7 -day windows), emphasizing the operational value of anomaly score monitoring in developing early-warning systems in WDNs.

Data records

The dataset is publicly available via Zenodo [20] (<https://doi.org/10.5281/zenodo.15096167>) under a Creative Commons Attribution 4.0 International (CC BY 4.0) license. The Zenodo record contains the following files:

- *input_model_potenc_predXfault7_A.csv* (CSV): primary dataset with 8,737 hourly records (timestamp, 18 anomaly-score variables, and the binary fault label *fault_d7*). Time coverage: 31-12-2021 23:00 to 30-12-2022 23:00 (Europe/Bratislava).
- *README.txt* (TXT): short description of the dataset, column meanings, license and contact details.

Folder structure: the Zenodo repository uses a flat structure; both files are located in the root directory.

The dataset's columns are categorized into three main categories: the column types included in the final dataset, along with their descriptions and data formats. All numerical variables are expressed as standardized anomaly scores (0–100) to ensure direct comparability across diverse sensor modalities and data sources.

Table 2. Structure and Semantics of Dataset Columns.

Column Type		Description	Format
timestamp		Date and time of measurement in hour resolution	DD-MM-YYYY HH:00 (time zone: Europe/Bratislava (CET/CEST))

fault_d7		Binary label: 1 = within ± 7 days of confirmed leak, 0 = otherwise	Integer (0/1)
*_kW, *_power_hour, *_kvar_hour		Anomaly scores from internal energy telemetry	Float (0–100)
*_ws_temp, *_ws_level, *_ws_vigor		Anomaly scores from internal water source sensors	Float (0–100)
*_anomaly_score		Environmental anomaly scores (e.g., temp, humidity, groundwater)	Float (0–100)

The higher values indicate greater deviation from the historical baseline of respective feature. Missing or offline data points are left empty (NULL) in the CSV to preserve data integrity.

While some features, such as groundwater level or hourly energy consumption, may not be available in all utilities, the dataset aims to represent a realistic pilot configuration based on accessible operational data rather than a globally standardized setup. Moreover, hydrometeorological data (e.g., groundwater level and precipitation) can often be accessed through cooperation with national or regional institutions that routinely monitor these indicators, and in the future, such data may also be collected directly by utilities using IoT-based sensors.

The dataset reflects a pilot-scale configuration based on data availability within the participating utility and does not include infrastructure design attributes such as pipe material or age.

Example Row Snapshot

Table 3 presents an illustrative subset of a single row from the final dataset to demonstrate its structure and content. Each row corresponds to one hourly timestamp and contains anomaly score values derived from multiple sensor domains, together with the binary fault label (*fault_d7*). The example highlights representative variables from energy telemetry, internal water source measurements, and environmental indicators.

Table 3. Example of a Single Row from the Final Dataset.

timestamp	fault_d7	...	4700008966_kW	4700008966_power_hour	279804_ws_level	gw_temp_site2_anomaly_score	...
...	...	...	...	...	...	...	...
15-07-2022 14:00	1	...	76.2	83.1	45.6	91.4	...
...	...	...	...	...	...	...	...

The complete dataset includes 18 selected predictor variables, grouped as follows:

- **Energy-related variables:** *4700008966_kW, 4700008966_power_hour, 4408411600_kW, 4408411600_power_hour, 4608927500_kvar_hour*
- **Internal water source metrics:** *238045_ws_temp, 279625_ws_temp, 279804_ws_level, 279805_ws_vigor, 279623_ws_vigor, 319235_ws_temp, 279831_ws_temp, 326417_ws_temp, 326416_ws_vigor*
- **Environmental indicators:** *temp_site1_anomaly_score, sfc_temp_site1_anomaly_score, gw_lvl_site2_anomaly_score, gw_temp_site2_anomaly_score*

All variables are expressed as standardized anomaly scores (0–100), enabling direct comparison across heterogeneous sensor types. This structure supports reproducibility, feature exploration, and unsupervised modeling within a consistent hourly time frame.

Technical Validation

To ensure the reliability, reproducibility, and scientific integrity of the dataset, multiple validation procedures were performed across data preprocessing and anomaly modeling phases:

- **Time-series integrity validation** were performed for all sensor streams to identify duplicate timestamps, discontinuities, and placeholder values (e.g., -9999). Sensors with less than 70% annual availability were excluded.
- **Anomaly model diagnostics** were conducted separately for each detection method. For Elastic ML, seasonal and diurnal decomposition residuals were visually analyzed to verify baseline stability and convergence. For Isolation Forest, sensitivity to the contamination parameter was evaluated and score stability was assessed using resampled subsets of environmental variables. All variables were transformed into probabilistic anomaly scores (0–100).
- **Cross-validation against ground-truth labels** was carried out using the Goodman–Kruskal gamma coefficient to quantify ordinal associations between anomaly scores and leak labels over ± 10 -day windows. Only variables with $\gamma \geq 0.10$ and $p < 0.05$ were retained. Only variables meeting predefined thresholds ($\gamma \geq 0.10$ and $p < 0.05$) were retained. Eighteen variables satisfied these criteria.
- **Temporal window validation** was performed by generating auxiliary fault labels using ± 1 to ± 10 -day windows around confirmed fault events. GLM (Adjusted univariate tests for repeated measure) and t-tests for independent samples were applied across all variables and time shifts (± 1 to ± 10 days). The ± 7 -day window (*fault_d7*) consistently provided a stable balance between temporal coverage and label variability, supporting its selection as the primary labeling strategy (binary fault label).
- **Transparency in missing data.** All NULL values were preserved for offline periods. This allows researchers to apply their own strategies for handling missing data.

Together, these technical validation steps ensure the statistical validity and analytical transparency of the dataset, making it suitable for downstream machine learning, fault prediction, and operational deployment studies.

Usage notes

The dataset is intended for research on smart water network monitoring and anomaly-based leak detection. It represents a one-year pilot configuration for a single water distribution zone and is designed as a reusable benchmark rather than a comprehensive representation of all WDN operating conditions.

The dataset is publicly available via Zenodo (<https://doi.org/10.5281/zenodo.15096167>) and consists of the primary CSV file (*input_model_potenc_predXfault7_A.csv*) containing 8,737 hourly records and 18

anomaly score variables together with a binary fault label (*fault_d7*), plus a README.txt file with column descriptions. All variables are aligned to a common hourly timestamp. Missing or offline sensor readings are left empty (NULL); no artificial imputation was applied and users should explicitly define their own missing-data handling strategies.

Key use cases:

- **Supervised learning:** The binary label *fault_d7* can be used to train classification models (e.g., random forests, gradient boosting, neural networks) for anomaly detection or fault prediction.
- **Unsupervised learning:** Researchers can cluster anomaly score vectors to identify recurring operational patterns or outliers that may indicate unknown failure modes.
- **Lead-lag analysis:** The ± 7 day windows and high-resolution timestamps enable temporal correlation studies, examining early warning signals relative to confirmed leaks.
- **Hybrid model construction:** since the dataset includes both internal (SCADA/energy) and external (environmental) indicators, it supports fusion-based models that integrate domain-specific and contextual predictors.
- **Sensor prioritization and interpretability:** Using feature importance scores (e.g., SHAP, permutation importance), researchers can evaluate the most predictive variables, optimizing sensor placement and model transparency.

Research implications:

This dataset provides a testbed for cross-domain feature integration and data fusion strategies, particularly in assessing the predictive value (power) of non-traditional variables. It is well-suited for comparative studies, allowing researchers to examine the added value of environmental signals alongside standard hydraulic telemetry. All interpretations presented in this manuscript are descriptive and limited to the characterization and validation of the dataset.

The dataset is provided in a ready-to-use format and can be analyzed directly or extended with user-defined preprocessing, modeling, or normalization strategies.

Support for segment-level modeling:

The dataset also enables **spatially segmented modeling of WDNs**: a) because each corresponds to a physical location or system component, facilitating location-specific model development, b) researchers can compare behavior across hydraulic zones, identifying differences in anomaly patterns, and c) the dataset supports benchmarking transferability between network segments, improving generalizability.

Integration guidance:

- All data are aligned on a common hourly timestamp for consistency.
- No artificial imputation was applied – NULL values should be handled explicitly during modeling.
- Anomaly scores are normalized (0–100), allowing direct cross-feature comparisons. If extended with new data, re-normalization may be necessary.

The absence of a *fault_d7* = 1 label does not imply leak-free operation. It only indicates that no confirmed fault was recorded in the OIS during that period. Consequently, precision-focused models should be evaluated accordingly.

Practical applications:

The dataset is intended to support methodological development and comparative evaluation in smart water distribution network monitoring. Its standardized anomaly score representation and hourly temporal resolution enable a range of analytical workflows, including:

- Leak detection and predictive maintenance benchmarking
- Evaluation of anomaly detection algorithms (including deep learning models)
- Lead-lag sensitivity studies for early warning systems
- Sensor optimization and feature importance analysis

- Transferability and zone-level model testing using multi-domain data
- Reproduction of the anomaly scoring process using the described workflow and tools

The dataset is provided in a ready-to-use format and can be directly analyzed or processed using user-defined scripts. Researchers can easily implement the described anomaly scoring and labeling methods or generate customized scripts using open-source tools or generative AI assistants to adapt the workflow to their specific data and research objectives.

Data Availability

The dataset described in this Data Descriptor is available in Zenodo at <https://doi.org/10.5281/zenodo.15096167> [20]. The public repository includes the primary CSV file (input_model_potenc_predXfault7_A.csv) with hourly anomaly scores and the fault_d7 leak-proximity label, along with a README.txt file describing the columns.

Code Availability

No custom source code is provided in the Zenodo repository associated with this dataset. The repository contains the finalized CSV file prepared for downstream analysis and model development. The procedures used to generate anomaly scores and fault labels are fully described in the Methods section of this manuscript. Because SCADA, energy, and environmental data differ across utilities in terms of sensor types, temporal resolution, system architecture, and data access constraints, users are encouraged to implement or adapt their own scripts for anomaly detection and labeling that are appropriate to their specific contexts.

The methods described can be reproduced using widely available tools and libraries, including Elastic Machine Learning (Elasticsearch X-Pack), scikit-learn (Isolation Forest), PyOD, and TensorFlow. In this study, Elastic Machine Learning was configured using a single-metric job with hourly resolution and daily and weekly seasonality, and Isolation Forest was implemented with $n_estimators = 100$, $max_features = 1.0$, $contamination = 0.05$, and $random_state = 42$, as detailed in the Methods.

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Author contributions

All authors contributed equally to the development of the dataset, methodology, analysis, and manuscript preparation. The work represents a collaborative effort across data engineering, statistical validation, domain expertise, and scientific writing.

Competing interests

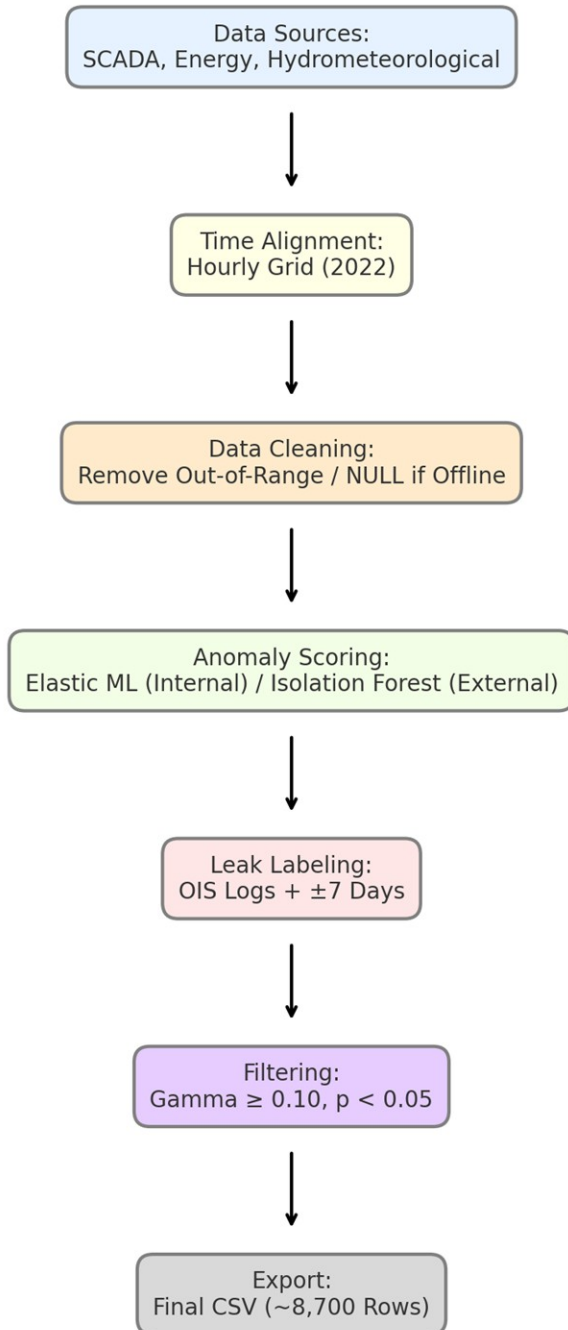
The authors declare no competing interests.

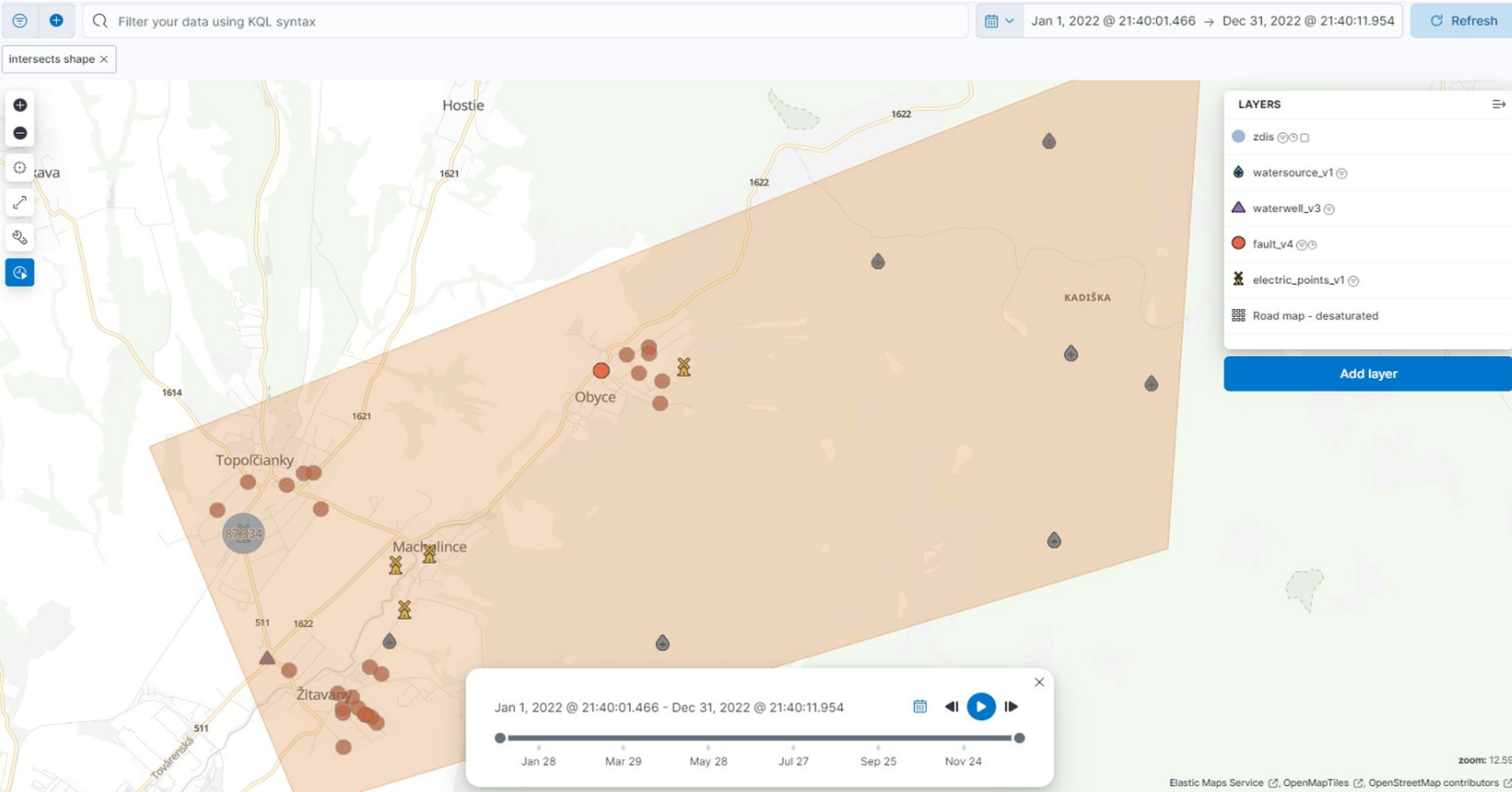
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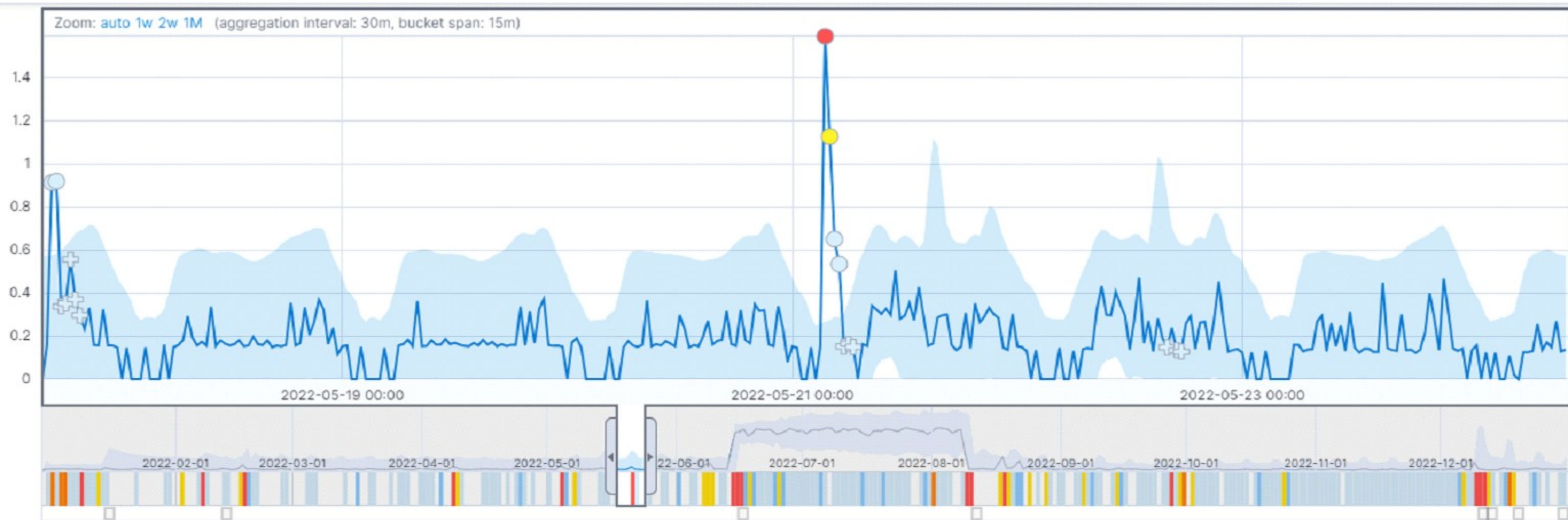
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Figure legends

- **Figure 1.** Schematic overview of the data collection and processing framework, illustrating integration of SCADA, energy, and environmental data into the final validated dataset.
- **Figure 2.** Map of the hydraulically isolated zone with sources, pump stations, and confirmed fault locations. The polygon delineates the study area.
- **Figure 3.** Example of a significant anomaly detected by Elastic ML in a pump station energy signal.
- **Figure 4.** The mean of Goodman-Kruskal gamma coefficients ($\pm$ standard deviation) across different time-shift windows (fault_d1 to fault_d10).
- **Figure 5.** This composite visualization shows time-series anomaly scores from 18 statistically validated features, grouped into three distinct domains:
 - (a) Energy-based variables (Elastic ML) exhibit sharp, high-amplitude peaks aligned with leak windows.
 - (b) Water source metrics (Elastic ML) show smoother fluctuations, indicating gradual hydraulic imbalance.
 - (c) Environmental indicators (Isolation Forest) demonstrate frequent, lower-intensity anomalies clustered around leak periods.



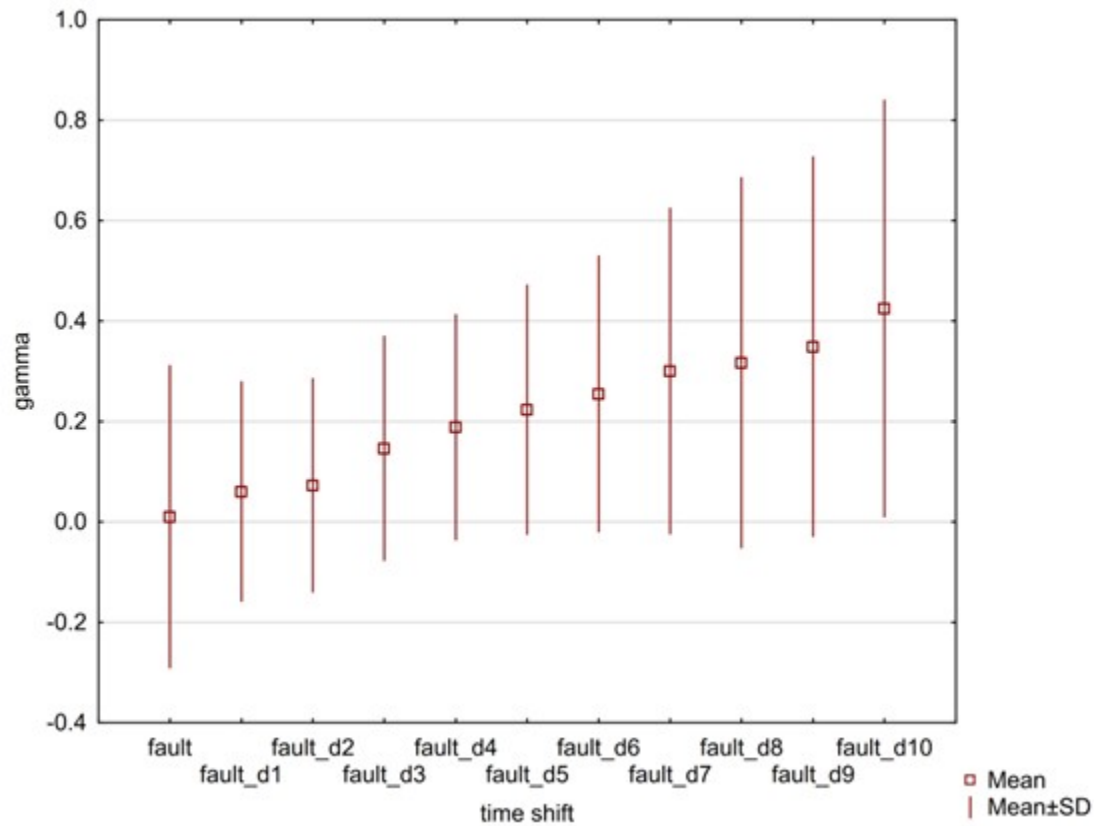




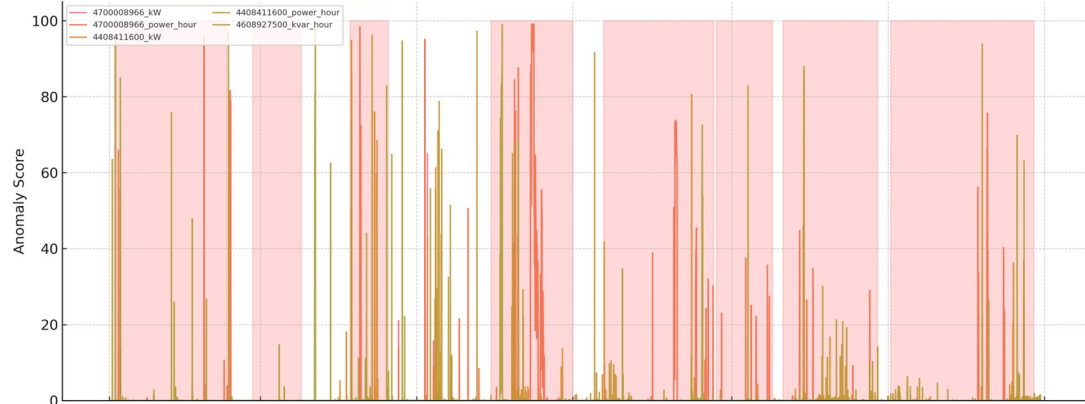
Anomalies

Severity ● warningInterval Auto

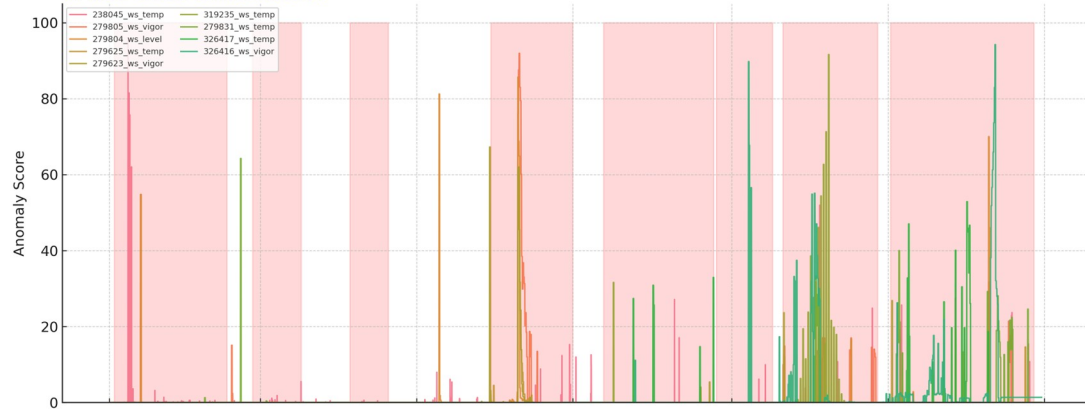
Time	Severity [?] ↓	Detector	Actual [?]	Typical [?]	Description	Actions
✓ May 21st 2022	● 79	mean(value)	2.432	0.02	↑ More than 100x higher	⚙



(a) Energy-Based Anomaly Scores [Elastic ML]



(b) Water Source Metrics [Elastic ML]



(c) Environmental Indicators [Isolation Forest]

